Miguel Ayala

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PROFESSIONAL SUMMARY

Software Engineer with a background in precision manufacturing, specializing in responsive, data-driven React and Next.js applications that merge technical precision with excellent user experience.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, HTML5, CSS3
Frameworks/Libraries: React, Next.js, Node.js, Express.js, Tailwind CSS
Databases & Query Languages: PostgreSQL, MySQL, MongoDB
Tools: Git, GitHub, GitLab, Docker, Linux, Bash
Core Competencies: Responsive Design, Accessibility (a11y), UI/UX, Web Performance, SEO Optimization

WORK EXPERIENCE

Software Engineer | Monarch Tractor | Livermore, CA
October 2023 – October 2025

- Built and maintained a full-stack web application using React, Node.js, and TypeScript to streamline data workflows, delivering a responsive, intuitive interface optimized for all devices.
- Implemented modern, responsive UI components with Tailwind CSS, standardizing design tokens and ensuring design consistency across breakpoints.
- Partnered with design and engineering teams to develop an interactive analytics dashboard (Next.js, SQLite, Recharts) that enabled real-time production monitoring and data-driven decision-making.
- Improved web performance and SEO through code-splitting, caching, and accessibility optimizations, achieving 90+ Core Web Vitals scores across key pages.

Metrology Technician | Monarch Tractor | Livermore, CA
April 2023 – September 2023

- Automated precision measurement workflows using PC-DMIS scripting, reducing inspection time 30% and improving process repeatability.
- Developed GD&T-driven inspection programs for complex geometries, enhancing accuracy and reducing manual intervention.
- Collaborated with engineering teams to improve data visualization and reporting interfaces, increasing accessibility of measurement data for cross-functional teams.

Hardware Test Engineer | Google | Mountain View, CA
January 2020 – March 2023

- Automated validation pipelines using Python-based data extraction and analysis tools integrated with CMM systems, accelerating FAI testing throughput.
- Developed Python scripts to generate automated SPC dashboards, cutting variance analysis time 40% and improving insights for hardware teams.
- Supported cross-functional R&D initiatives, bridging hardware data outputs with software-driven reporting solutions to ensure accuracy and efficiency.

Previous Experience | Various Companies
May 2011 – December 2019

- Quality assurance and precision measurement roles in aerospace manufacturing, specializing in inspection automation, GD&T interpretation, and non-conformance reduction.

EDUCATION

Bachelor of Science in Computer Science | Western Governors University | Salt Lake City, UT
April 2025
Master of Science in Computer Science | Georgia Institute of Technology | Atlanta, GA
Expected May 2029